WINTER SEMESTER 2025-26

ASSESSMENT - 1

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School: School of Computer Science Engineering and Information Systems (SCORE)

Course Code: ISWE406P

Course Title: Agile Development Process and Devops Lab

Faculty: Dr. Sweta Bhattacharya

Slot: L49+L50

CASE STUDIES

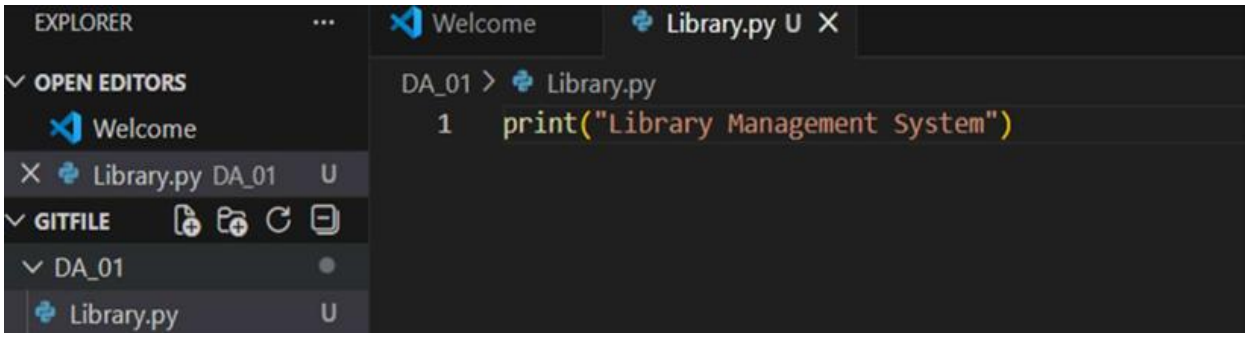
Use appropriate Git Commands to Execute the following case studies in VS Code

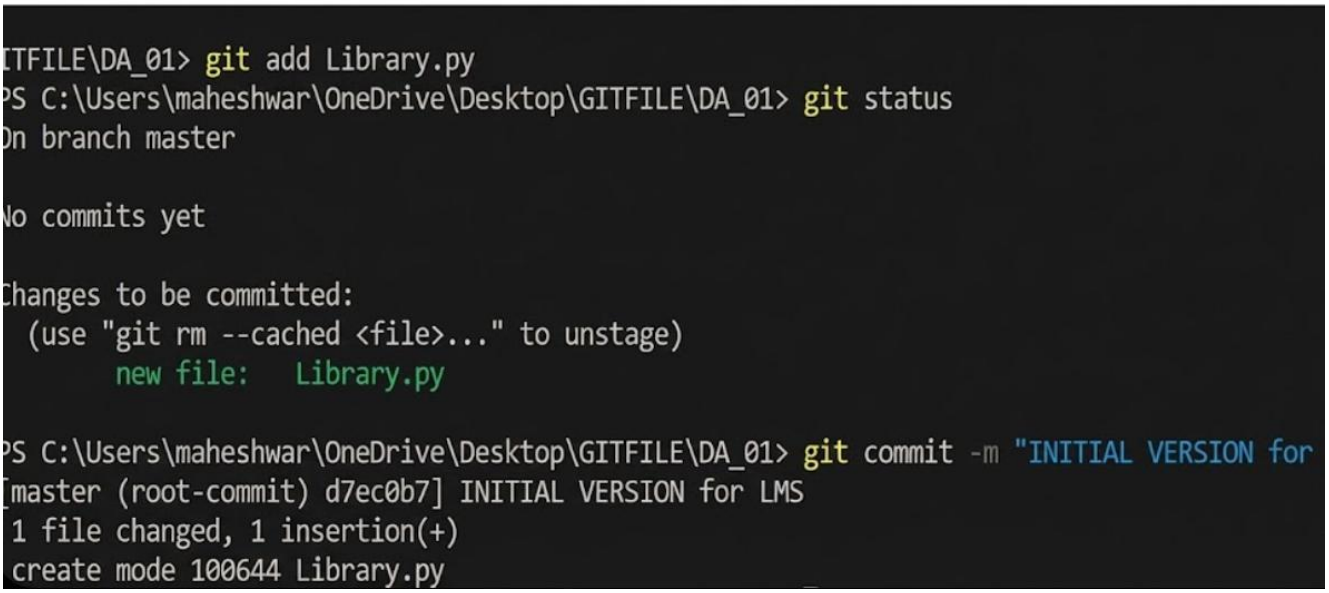
Case Study 1: New Project Version Control Setup

Scenario: You are assigned to develop a Library Management System. Before coding, your team decides to enable version control.

Tasks:

- Create a project folder and open it in VS Code
- Initialize a Git repository
- Create a Python file and track it using Git
- Commit the initial version



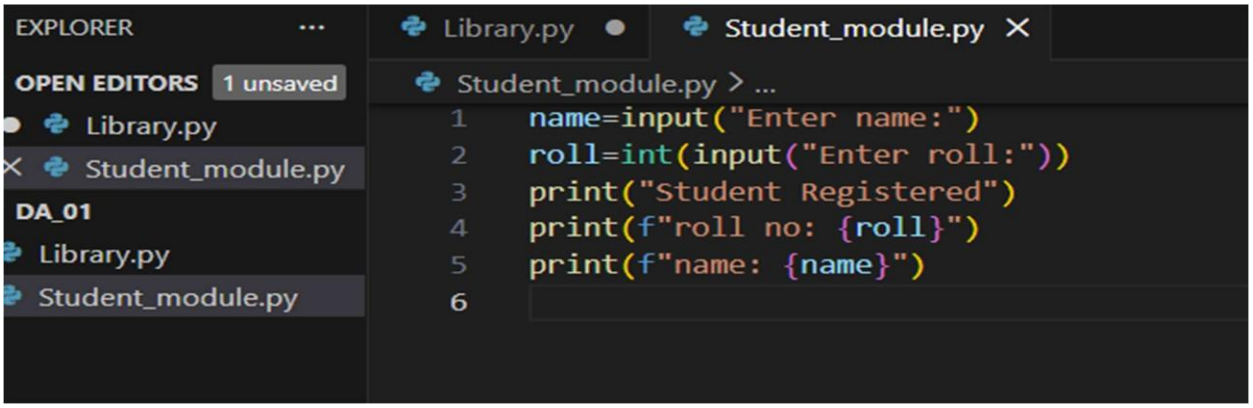


Case Study 2: Tracking Daily Code Changes

Scenario: You are working on a Student Registration Module. You make small updates daily and need to record them.

Tasks:

- Modify an existing file
- Check file differences
- Stage and commit changes
- View commit history




```
PS C:\Users\maheshwar\Dive\Desktop\GITFILE\DA_01> git init
Inititialized existing Git repository in C:/Users/pursh/OneDrive/Desktop/GITFILE/DA_01/.git/
PS C:\Users\maheshwar\Dive\Desktop\GITFILE\DA_01> git diff
PS C:\Users\pu   \OneDrive\Desktop\GITFILE\DA_01>
PS C:\Users\maheshwar\Dive\Desktop\GITFILE\DA_01> git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Student_module.py

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\maheshwar\Dive\Desktop\GITFILE\DA_01> git add Student_module.py
PS C:\Users\maheshwar\Dive\Desktop\GITFILE\DA_01> git commit -m "Remod Student module"
[master 3a1511f] Remod Student module
 1 file changed, 5 insertions(+)
 create mode 100644 Student_module.py
PS C:\Users\maheshwar\Dive\Desktop\GITFILE\DA_01> git status
On branch master
nothing to commit, working tree clean
PS C:\Users\maheshwar\Dive\Desktop\GITFILE\DA_01> git log --oneline
3a1511f (HEAD -> master) Remod Student module
d7ec0b7 INITIAL VERSION for LMS
PS C:\Users\maheshwar\Dive\Desktop\GITFILE\DA_01>
```

Case Study 3: Feature Development Using Branches

Scenario:

A new Login feature must be developed without disturbing main code.

Tasks:

- Create a new branch
- Add login-related code
- Commit changes
- Switch between branches

```
PS C:\Users\maheshwar\Desktop\GITFILE\DA_01> git branch
Reinitalized existingGit repository in
PS C:\Users\maheshwar\Desktop\GITFILE\DA_01> git checkout -b "Login-feature"
Switched to a new branch 'Login-feature'
PS C:\Users\maheshwar\Desktop\GITFILE\DA_01> git branch
* Login-feature
  master
PS C:\Users\maheshwar\Desktop\GITFILE\DA_01> git add Student_module.py
PS C:\Users\maheshwar\Desktop\GITFILE\DA_01> git commit -m "Added login-feature"
Login-feature 502b43a] Added login-feature
1 file changed, 9 insertions(+), 5 deletions(-)
PS C:\Users\maheshwar\mahestop\GITFILE\DA_01> git checkout master
Switched to branch 'master'
PS C:\Users\maheshwar\maheshap\GITFILE\DA_01> git checkout "Login-feature"
Switched to branch 'Login-feature'
PS C:\Users\maheshwar\maheshap\GITFILE\DA_01>
```

Case Study 4: Merging Feature into Main Code

Scenario: The login feature is tested and approved.

Tasks:

- Switch to main branch
- Merge feature branch
- Verify merged content

```
PS C:\Users\maheshwar\OneDrive\ \GITFILE\DA_01> git init
Reitialized exsting Git repository in C:/Users\war\OneDrive\Desktop\GITFILE/DA_01.git/
PS C:\Users\maheshwar\OneDrive\ \GITFILE\DA_01> git diff
Switched to a new branch 'Login-feature'
PS C:\Users\maheshwar\OneDrive\ \GITFILE\DA_01> git branch
* Login-feature
  master
PS C:\Users\maheshwar\OneDrive\ \GITFILE\DA_01> git add Student_module.py
PS C:\Users\maheshwar\OneDrive\ \GITFILE\DA_01> git commit -m "Added login-feature"
[Login-feature 502b43a] Added login-feature
 1 file changed, 9 insertions(+), 5 deletions(-)
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\ DA_01> git checkout master
Switched to branch 'master'
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\ DA_01> git merge Login-feature
Already up to date.
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\ DA_01>
```

Case Study 5: Working with Remote Repository

Scenario: The project must be shared using GitHub.

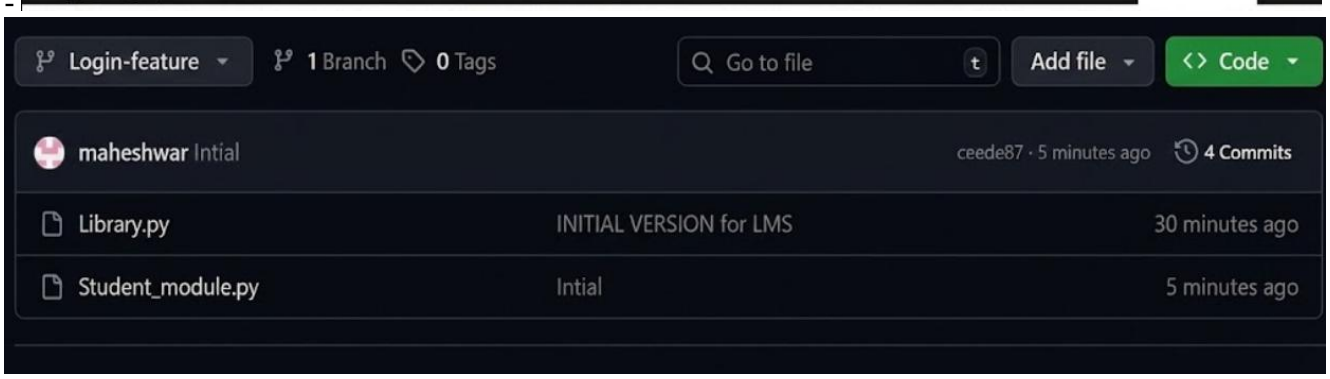
Tasks:

- Create remote repository
- Add remote origin
- Push code to GitHub

```
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01>git remote add origin https://github.com/maheshwarrg2005/DA-01.git
error: remote origin already exists.
```

```
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01>git remote -v
origin  https://github.com/maheshwar/maheswar/DA-01.git (fetch)
origin  https://github.com/maheshwar/maheswar/DA-01.git (push)
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git push -u origin Login-feature
branch 'Login-feature' set up to track 'origin/Login-feature'.
Everything up-to-date
```

-



Case Study 6: Undoing Mistakes

Scenario: A file is modified incorrectly.

Tasks:

- View file changes
- Discard uncommitted changes

```
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git status
On branch master
nothing to commit, working tree clean
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git diff
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git restore Student_module.py
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01>
```

```
Student_module.py > ...                    C:\Users\
1   name=input("Enter name:")
2   roll=int(input("Enter roll:"))
3   print("Student Registered")
4   print(f"roll no: {roll}")
5   print(f"name: {name}")
6
```

Case Study 7: Handling Merge Conflicts

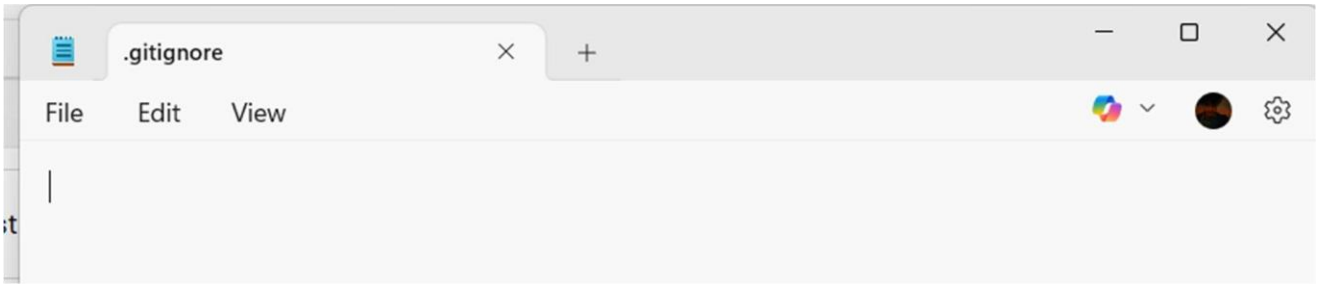
Scenario: Two developers edited the same file.
Tasks:
- Merge branches
- Resolve conflicts
- Commit resolved changes

```
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git checkout master
Already on 'master'
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git merge Login-feature
Updating 3a1511f..ceede87
Fast-forward
 Student_module.py | 25 ++++++++++++++++++++-----
 1 file changed, 20 insertions(+), 5 deletions(-)
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git add merge.py
fatal: pathspec 'merge.py' did not match any files
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git add merge.py
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git commit -m "Handled conflict while merging"
[master 34e163a] Handled conflict while merging
 1 file changed, 1 insertion(+)
 create mode 100644 merge.py
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git status
On branch master
nothing to commit, working tree clean
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01>
```

Case Study 8: Ignoring Unwanted Files

Scenario: Log and temporary files should not be tracked.
Tasks:
- Create .gitignore file
- Add ignore patterns

```
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> notepad .gitignore
```




```
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git add .gitignore
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git commit -m "Added .gitignore to ignore log and temp fi
[master 65cd94a] Added .gitignore to ignore log and temp files
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 .gitignore
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01>
```

Case Study 9: Collaborating with Team

Scenario: A teammate pushed new changes.
Tasks:
- Fetch latest changes
- Pull updates
- Resolve conflicts if any

```
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git fetch origin
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git pull origin main
fatal: couldn't find remote ref main
3a1511f Remod Student module
d7ec0b7 INITIAL VERSION for LMS
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git pull origin
You asked to pull from the remote 'origin', but did not specify
a branch. Because this is not the default configured remote
for your current branch, you must specify a branch on the command line.
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git pull origin main
fatal: couldn't find remote ref main
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git pull origin master

fatal: couldn't find remote ref master
```

Case Study 10: Final ProjectRelease

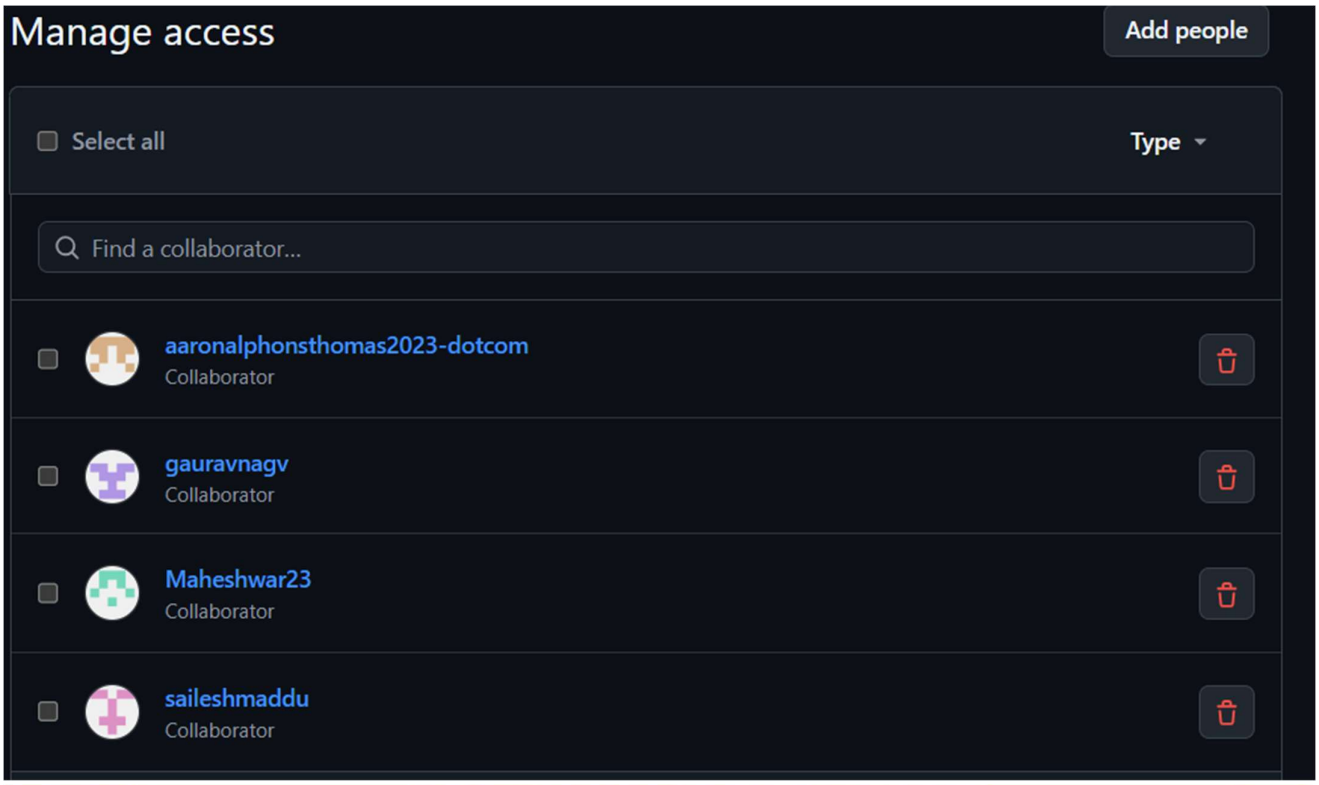
Scenario: Project is ready for final submission.
Tasks:
- Review commit history
- Push final code
- Prepare release version

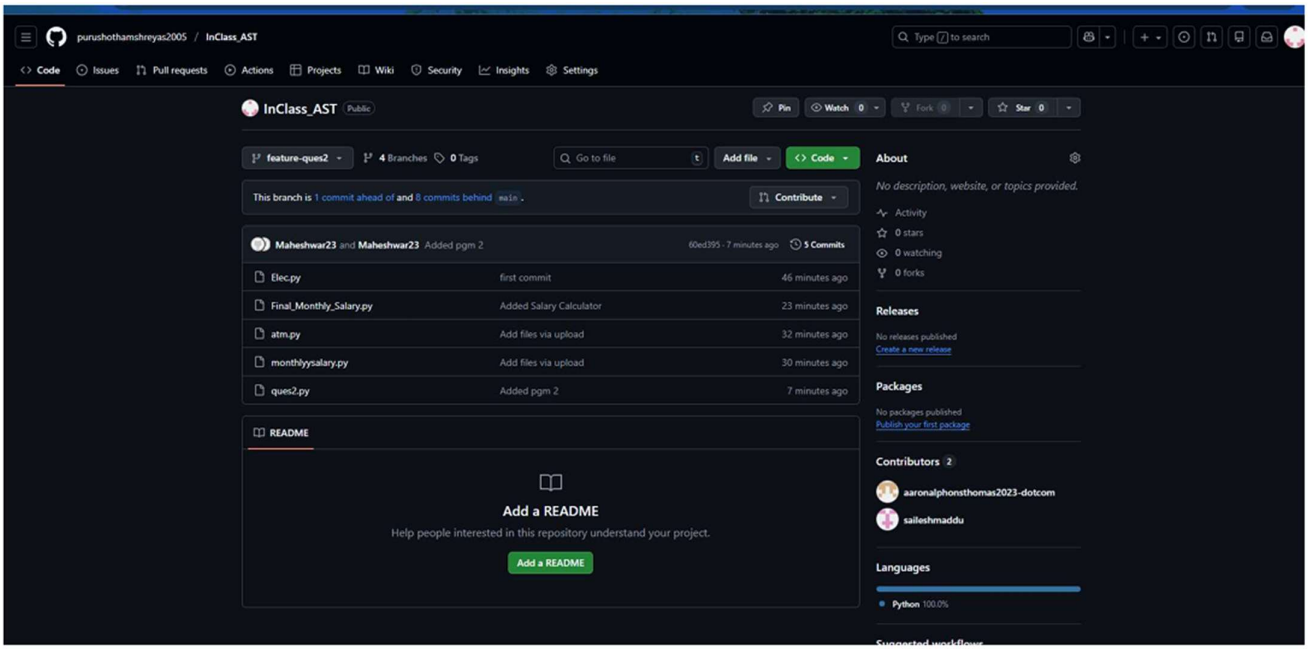
```
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git log --oneline
65cd94a (HEAD -> master) Added .gitignore to ignore log and temp files
34e163a Handled conflict while merging
ceede87 (origin/Login-feature, origin/HEAD, Login-feature) Intial
502b43a Added login-feature
3a1511f Remod Student module
d7ec0b7 INITIAL VERSION for LMS
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git status
On branch master
nothing to commit, working tree clean
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git checkout master
Already on 'master'
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01> git push origin master
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 611 bytes | 203.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:      https://github.com/maheshwar2005/DA-01/pull/new/master
remote:
To https://github.com/maheshwar2005/DA-01.git
 * [new branch]      master -> master
PS C:\Users\maheshwar\OneDrive\Desktop\GITFILE\DA_01>
```

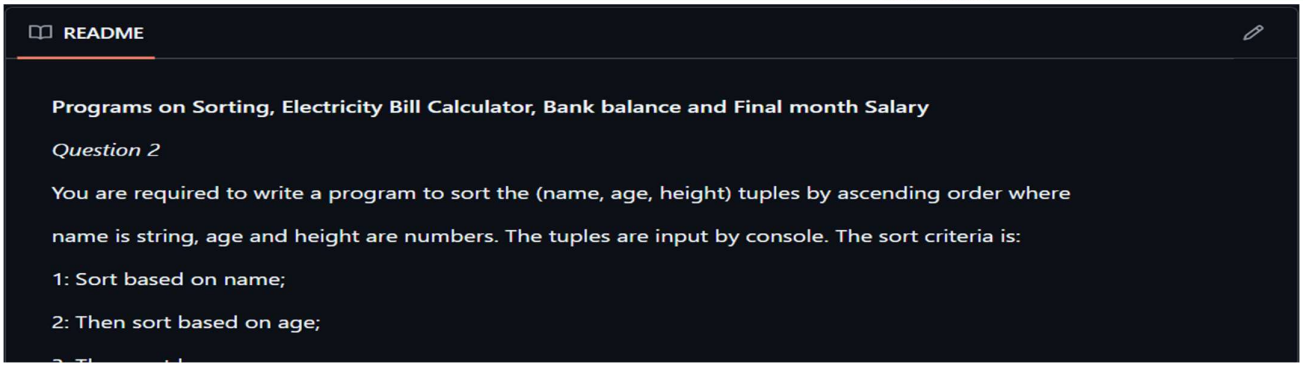
***

IN-CLASS ASSIGNMENT

Collaboration in GitHub:







Question 2. You are required to write a program to sort the (name, age, height) tuples by ascending order where name is string, age and height are numbers. The tuples are input by console. The sort criteria is: 1: Sort based on name; 2: Then sort based on age; 3: Then sort by score.

The priority is that name > age > score.

If the following tuples are given as input to the program:

Tom,19,80 John,20,90 Jony,17,91 Jony,17,93 Json,21,85

Then, the output of the program should be:

[('John', '20', '90'), ('Jony', '17', '91'), ('Jony', '17', '93'), ('Json', '21', '85'), ('Tom', '19', '80')]

```
PS C:\Users\batch1> git clone https://github.com/purushothamshreyas2005/InClass_AST.git
fatal: destination path 'InClass_AST' already exists and is not an empty directory.
PS C:\Users\batch1> cd InClass_AST
PS C:\Users\batch1\InClass_AST> git add second.py
fatal: pathspec 'second.py' did not match any files
PS C:\Users\batch1\InClass_AST> code .
PS C:\Users\batch1\InClass_AST>
```

```
PS C:\Users\batch1\InClass_AST> git add .
PS C:\Users\batch1\InClass_AST> git commit -m "Added pgm 2"
[main 60ed395] Added pgm 2
 Committer: Maheshwar23 <Batch1@VITUNIVERSITY.LOCAL>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author
```

```
PS C:\Users\batch1\InClass_AST> git branch feature-ques2
PS C:\Users\batch1\InClass_AST> git push -u origin feature-ques2
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 24 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 649 bytes | 649.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature-ques2' on GitHub by visiting:
remote:      https://github.com/purushothamshreyas2005/InClass_AST/pull/new/feature-ques2
remote:
```

```
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 24 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 649 bytes | 649.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature-ques2' on GitHub by visiting:
remote:      https://github.com/purushothamshreyas2005/InClass_AST/pull/new/feature-ques2
remote:
To https://github.com/purushothamshreyas2005/InClass_AST.git
 * [new branch]      feature-ques2 -> feature-ques2
branch 'feature-ques2' set up to track 'origin/feature-ques2'.
PS C:\Users\batch1\InClass_AST>
```

Question 3: An electricity board calculates charges based on usage. Write a program that: • Takes number of units consumed • Calculates bill: o First 100 units → ₹2/unit o Next 100 units → ₹3/unit o Remaining → ₹5/unit • Displays total bill

Welcome | Elec.py

Elec.py > ...

```python

units = int(input("Enter the number of units consumed: "))

bill = 0

if units <= 100:

    bill = units * 2
elif units <= 200:

    bill = (100 * 2) + (units - 100) * 3
else:

    bill = (100 * 2) + (100 * 3) + (units - 200) * 5


print(f"Total electricity bill: ₹{bill}")
```

```
PS C:\Users\batch1\Desktop\AGILEE> & "C:/Program Files/Python310/python.exe" c:/Users/batch1/Desktop/AGILEE/Elec.py
Enter the number of units consumed: 100
Total electricity bill: ₹200
PS C:\Users\batch1\Desktop\AGILEE> 500
500
PS C:\Users\batch1\Desktop\AGILEE> git add Elec.py
PS C:\Users\batch1\Desktop\AGILEE> git commit -m "first commit"
[main (root-commit) e4535c7] first commit
 Committer: Batch1 <Batch1@VITUNIVERSITY.LOCAL>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 17 insertions(+)
 create mode 100644 Elec.py
PS C:\Users\batch1\Desktop\AGILEE> git branch -M main
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git push -u origin main
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git push -u origin main
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git push -u origin main
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git push -u origin main
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git push -u origin main
info: please complete authentication in your browser...
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git push -u origin main
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git remote add origin https://github.com/purushothamshreyas2005/InClass_AST.git
PS C:\Users\batch1\Desktop\AGILEE> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 24 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 382 bytes | 382.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/purushothamshreyas2005/InClass_AST.git
```

Question 4: An ATM machine allows withdrawals only if sufficient balance is available. Write a program that: • Stores a user's balance • Takes withdrawal amount as input • Checks: o Amount must be a multiple of 100 o Balance must be sufficient • Deducts amount and displays updated balance

```
Id CommandLine
-- -----------
 1 try { . "c:\Program Files\Microsoft VS Code\resources\app\out\vs\workbench\contrib\terminal\common\scripts\shellInte...
 2 git clone https://github.com/purushothamshreyas2005/InClass_AST.git
 3 git add atm.py
 4 git init
 5 git add atm.py
 6 git commit -m
 7 git commit -m "atm program"
 8 git branch -M main
 9 git remote add origin https://github.com/purushothamshreyas2005/inClass_AST.git
10 git push -u origin main
```

Question 5: A company calculates monthly salary based on basic pay. Write a program that: • Accepts basic salary • Calculates: o HRA = 20% of basic o DA = 10% of basic o Tax = 5% of total salary • Displays net salary

```
PS C:\Users\batch1> git clone https://github.com/purushothamshreyas2005/InClass_AST.git
Cloning into 'InClass_AST'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 9 (delta 1), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
Resolving deltas: 100% (1/1), done.
PS C:\Users\batch1> cd InClass_AST
PS C:\Users\batch1\InClass_AST> code .
PS C:\Users\batch1\InClass_AST>
```

```
EXPLORER
INCLASS_AST
atm.py
Elec.py
Final_Monthly_Salary....
monthlyysalary.py

Welcome | Final_Monthly_Salary.py
Final_Monthly_Salary.py > ...
1  print("SALARY GENERATING")
2  sal=int(input("Enter basic salary:"))
3  hra=0.20*sal
4  da=0.10*sal
5  tot=sal+hra+da
6  tax=0.05*tot
7  net=tot+tax
8  print("Total salary is",net)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\batch1\InClass_AST> git add .
PS C:\Users\batch1\InClass_AST> git commit -m "Added Salary Calculator"
[main e1e2e22] Added Salary Calculator
 Committer: Batch1 <Batch1@VITUNIVERSITY.LOCAL>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit
    git config --global --edit
    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 8 insertions(+)
 create mode 100644 Final_Monthly_Salary.py
PS C:\Users\batch1\InClass_AST> git branch calc-branch
PS C:\Users\batch1\InClass_AST> git push -u origin calc-branch
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 24 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 489 bytes | 489.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'calc-branch' on GitHub by visiting:
remote:      https://github.com/purushothamshreyas2005/InClass_AST/pull/new/calc-branch
remote:
To https://github.com/purushothamshreyas2005/InClass_AST.git
 * [new branch]      calc-branch -> calc-branch
branch 'calc-branch' set up to track 'origin/calc-branch'.
PS C:\Users\batch1\InClass_AST> git branch -M main
PS C:\Users\batch1\InClass_AST> git push -u origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/purushothamshreyas2005/InClass_AST.git
   d0fea55..e1e2e22  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\batch1\InClass_AST>

OUTLINE
TIMELINE
```

***